

WHAT DRIVES CRYPTOCURRENCY RETURNS? A SPARSE STATISTICAL JUMP MODEL APPROACH

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ABSTRACT. We apply the statistical sparse jump model, a recently developed, interpretable and robust regime switching model, to infer key features that drive the return dynamics of the largest cryptocurrencies.

The algorithm jointly performs feature selection, parameter estimation, and state classification. Our large set of candidate features are based on cryptocurrency, sentiment and financial market-based time series that have been identified in the emerging literature to affect cryptocurrency returns, while others are new.

In our empirical work, we demonstrate that a three-state model best describes the dynamics of cryptocurrency returns. The states have natural market-based interpretations as they correspond to bull, neutral, and bear market regimes, respectively. Using the data-driven feature selection methodology, we are able to determine which features are important and which ones are not. In particular, out of the set of candidate features, we show that first moments of returns, features representing trends and reversal signals, market activity and public attention are key drivers of crypto market dynamics.

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Key words and phrases. Clustering; Blockchain; Cryptocurrencies; Feature Selection; Regime Switching; Unsupervised Learning.

1. INTRODUCTION

Since the introduction of Bitcoin in 2009, which operates with a decentralized ledger system known as the blockchain, cryptocurrencies have attracted much attention, and today some market participants argue that they constitute a separate asset class (Bianchi, 2020; Pele et al., 2021). Following the introduction of Bitcoin, many other cryptocurrencies have been implemented, collectively referred to as altcoins. By March 2022, there were approximately 18K cryptocurrencies in total.

Time series analysis of cryptocurrency dynamics show they exhibit regime switching, structural breaks and jumps in both returns and volatility, see Figure 1 (Ardia et al., 2019; Chaim et al., 2018; Shen et al., 2020). Later, Rydén et al. (1998) demonstrate empirically that a simple *hidden Markov model* (HMM) can reproduce most of the common stylized facts in asset returns (cf. Cont (2001) and Lindström et al. (2015, Chapter 1)). Koki et al. (2022) estimate several different HMMs for the purpose of forecasting Bitcoin, Ethereum and Ripple and demonstrate that a four-state specification provides the best out-of-sample performance. However, the statistical properties of the hidden states are not consistent across the three cryptocurrencies, making an interpretation of the latent states as different economic regimes difficult. An alternative explanation to their empirical results is that the switching dynamics is governed by some exogenous process(es) not included in their models.

While the understanding of the key drivers of cryptocurrency returns is still emerging, the literature has identified a number of crypto-, sentiment- and financial market-related features that impact cryptocurrency market dynamics (see, for example Koki et al. (2022) and Yae et al. (2022) for an overview). Kristoufek (2015) finds that trade and transaction volumes are positively correlated with Bitcoin returns. Similarly, Aalborg et al. (2019) suggest that frequent use of Bitcoin would increase its demand, but also that this relationship might be negative during periods of high volatility. Catania et al. (2019) and Bianchi (2020) study the dependence of many cryptocurrencies on macroeconomic factors like S&P500 returns, market volatility (VIX) and commodities. They find a weak correlation between returns on cryptocurrencies and commodities, especially gold. Yae et al. (2022) demonstrate that the change in correlation between Bitcoin and S&P500 returns predicts future Bitcoin returns. In particular, they show that an increase (decrease)

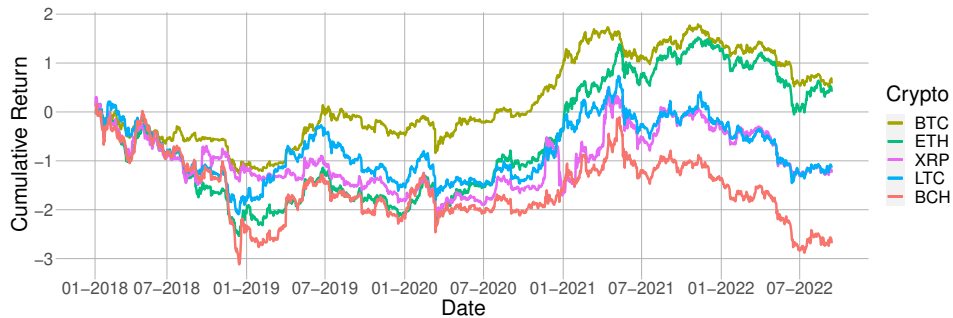


FIGURE 1. Cumulative log-returns of the five cryptocurrencies Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), Litecoin (LTC), and Bitcoin Cash (BCH), over the period January 2018 - September 2022.

in correlations today suppresses (boosts) Bitcoin prices the following day. Xiong et al. (2020) provide evidence of the importance of including production cost and use value (a measure of how many people use a particular coin as a mean of exchanging value) of Bitcoin, as measured by the number of unique addresses and the Bitcoin hash rate. Liu et al. (2021) and Liu et al. (2022) find that cross-sectional factors constructed from market return, size, momentum, and public attention can forecast cryptocurrency returns. Cheah et al. (2020) and Koki et al. (2022) also find that time series momentum is a statistically significant predictor for the upward and downward trending states of crypto markets.

In his seminal work, Hamilton (1989) suggests that the dynamics of financial returns can be described by Markovian regime-switching processes, with drastic breaks associated with events like economic crises or political events.

In this article, we aim to determine what are the most important drivers of the return dynamics of the five largest and most liquid cryptocurrencies; namely, Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), Litecoin (LTC), and Bitcoin Cash (BCH). For this purpose, we construct a large set of candidate features grouped into three categories: financial market, sentiment and crypto market-related features. While many of these features are drawn from the emerging literature, we also propose some that are new.

We adopt the *sparse statistical jump model* (SJM) of Nystrup et al. (2021) to select the features that best explain cryptocurrency returns. As an alternative modeling framework to HMMs, the SJMs have several advantages.

First, they allow us to *simultaneously* perform parameter estimation, state-sequence decoding and feature selection. Second, Nystrup et al. (2021) show that, in contrast to many “competing” models, SJMs are more robust to misspecification and initialization, deliver acceptable performance even on smaller samples, and tend to be more efficient for high-dimensional feature vectors. Third, the estimated SJMs are easy to interpret from their conditional state dynamics and the weight associated with each feature, providing an opportunity to reconcile statistical properties and selected features with observed market behavior and economic intuition.

We demonstrate in our empirical study that when more than four hundred features are included, a three-state SJM best explains cryptocurrency returns. The three states have intuitive economic interpretations, corresponding to bull, neutral, and bear market regimes. The relevant features selected by the SJM are exponential moving averages of returns, features representing trend and reversal signals drawn from the technical analysis literature, market activity and public attention. The features that we use for representing market activity and public attention are new.

The rest of the article is organized as follows. In Section 2 we review the mathematical formulation of SJMs, their implementation and hyperparameter tuning. In Section 3 we describe the candidate features we use in inferring drivers of cryptocurrency returns. We describe the preparation of our dataset and present the results from our empirical analysis in Section 4. Section 5 concludes. Appendix A provides a complete list of all the features we use in this study.

2. METHODOLOGY

Traditional regime switching models, such as HMMs that have been used for decades in finance and economics, strike a balance between being interpretable and flexible enough to model complex non-stationary behavior. Well-known limitations of this class of models include sensitivity to model misspecification and feature selection for exogenous variables (see Zucchini et al. (2017) for a recent overview). Another issue is the difficulty to reliably estimate model parameters as the log-likelihood function is notoriously multimodal. Estimation frameworks such as direct maximization of the log-likelihood, the expectation-maximization algorithm and Bayesian approaches are compared in Rydén (2008), with no uniform preference of any particular framework over the others.

Recently, Bemporad et al. (2018) introduced the class of *statistical jump models* (JM), nesting e.g. HMMs. We remark that JMs are not related to jump-diffusion models, a common class of stochastic processes. Following the trend in statistics and machine learning of reformulating probabilistic models as well-behaved optimization problems, such as LASSO or support vector machines, a JM is estimated by minimizing the combined loss of the (negative) likelihood and penalties for jumping between states.

Nystrup et al. (2020b) reformulate the JM as a temporal clustering model. They show in a simulation study that the JM interpretation of an HMM has many advantages, including robustness against model misspecification and poor initialization of the optimizer, as well as surprisingly rapid convergence, typically within only a few iterations. Nystrup et al. (2020b) proves that the global loss can be optimized by sequentially alternating between (a) fitting the model parameters while keeping the state sequence fixed, and (b) estimating the state sequence through dynamic programming while keeping the model parameters fixed.

In a further extension, Nystrup et al. (2021) introduce the *sparse jump model* (SJM) by incorporating feature selection from the clustering literature. This is important as large scale feature selection has been infeasible for standard HMMs. They demonstrate in a series of simulation studies that SJMs outperform competing frameworks and are capable of recovering true features while rejecting false features with high probability even when considering hundreds of them.

2.1. Mathematical Formulation of the SJM. The JM proposed by Bemporad et al. (2018) is governed by a latent state sequence, $\{s_t\}$, switching between K states, that each is associated with a vector of parameters, $\{\theta_k\}_{k=1}^K$. The model is fitted to a time series $\mathbf{y}_1, \dots, \mathbf{y}_T$, by minimizing the loss

$$\sum_{t=1}^{T-1} [\ell(\mathbf{y}_t; \theta_{s_t}) + \lambda \mathbb{I}_{\{s_t \neq s_{t-1}\}}] + \ell(\mathbf{y}_T; \theta_{s_T}), \quad (1)$$

where the hyperparameter $\lambda \geq 0$ controls the number of jumps between states and $\ell(\cdot)$ is some loss function to be specified.

Nystrup et al. (2020b) consider temporal clustering by transforming the data into P standardized features, $\tilde{\mathbf{y}}_{t,p} \in \mathbb{R}^P$, and using a quadratic loss

function. Their corresponding objective function is given by

$$\sum_{t=1}^{T-1} [\|\tilde{\mathbf{y}}_{t,p} - \boldsymbol{\mu}_{s_t}\|_2^2 + \lambda \mathbb{I}_{\{s_t \neq s_{t-1}\}}] + \|\tilde{\mathbf{y}}_{T,p} - \boldsymbol{\mu}_{s_T}\|_2^2, \quad (2)$$

where $\boldsymbol{\mu}_{s_t}$ is the conditional mean of state s_t . It can be shown that this form of temporal clustering collapses into K -means clustering when $\lambda = 0$.

Nystrup et al. (2021) note that the *total sum of squares* (TSS) can be expressed as the sum of *within-cluster sum of squares* (WCSS) and the *between cluster sum of squares* (BCSS)

$$\sum_{t=1}^T \|\tilde{\mathbf{y}}_{t,p} - \bar{\boldsymbol{\mu}}\|_2^2 = \sum_{t=1}^T \|\tilde{\mathbf{y}}_{t,p} - \boldsymbol{\mu}_{s_t}\|_2^2 + \sum_{k=1}^K n_k \|\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}}\|_2^2, \quad (3)$$

where $\bar{\boldsymbol{\mu}}$ is the unconditional mean of the features and $\boldsymbol{\mu}_k$ the conditional mean of the features in the k -th state.

Building upon the work by Witten et al. (2010) on feature selection in clustering, and using that minimizing the WCSS in equation (2) is equivalent to maximizing the BCSS (as the TSS is constant), Nystrup et al. (2021) propose to solve

$$\begin{aligned} \max_{\boldsymbol{\mu}_k, \{s_t\}, \mathbf{w}} \quad & \mathbf{w}' \sum_{k=1}^K n_k (\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}})^2 - \lambda \sum_{t=1}^{T-1} \mathbb{I}_{\{s_t \neq s_{t-1}\}} \\ \text{subject to} \quad & \|\mathbf{w}\|^2 \leq 1, \quad \|\mathbf{w}\|_1 \leq \kappa \\ & w_p \geq 0 \quad \forall p, \end{aligned} \quad (4)$$

where $\mathbf{w}$ is the vector of feature weights, n_k denotes the number of observations belonging to the k -th cluster and the hyperparameter $1 \leq \kappa \leq \sqrt{P}$ controls the degree of sparsity of the features. Nystrup et al. (2021) show that this optimization problem can be solved by iteratively alternating between (a) fitting model parameters given the state sequence, $\{s_t\}$, and weights $\mathbf{w}$, (b) deriving the state sequence, $\{s_t\}$, by solving a dynamic program, essentially running the Viterbi algorithm backwards, and (c) updating the weights $\mathbf{w}$ using soft thresholding.

2.2. Model Implementation and Hyperparameters. As the SJM is unsupervised, we select its hyperparameters based on a version of the *generalized information criteria* (GIC) (Fan et al., 2013) for high-dimensional, penalized models suitably modified for SJMs.

Fan et al. (2013) consider a setup where the number of features is considerably larger than the number of observations. They define a GIC as

$$\text{GIC} = \frac{1}{T} \left\{ 2 \left(\ell_T(Y, Y) - \ell_T(\hat{\theta}_A, Y) \right) + a_T M \right\}, \quad (5)$$

where M is a measure of model complexity, a_T is a penalty term possibly depending on the number of observations T and features P , $\ell_T(\hat{\theta}_{\mathcal{A}}, Y)$ is the log-likelihood computed using the estimated active set of features, $\mathcal{A}$, and $\ell_T(Y, Y)$ is the log-likelihood for the saturated model, which is the model obtained considering the entire set of features. Fan et al. (2013) note that the generalized versions of the *Akaike's information criteria* (AIC) (Akaike, 1974) or the *Bayesian information criteria* (BIC) (Schwarz, 1978) are recovered by setting M equal to the total number of parameters, and $a_T = 2$ or $a_T = \log(T)$ for AIC and BIC, respectively.

To construct a proper index, we derive expressions for the model fit and complexity and find that a BIC adjusted for SJM is remarkably efficient in selecting the correct hyperparameters values (see, Cortese et al. (2023), for details), while the Fan et al. (2013) criteria is slightly less efficient. The AIC is designed to find the best predictive model, but does not consistently select the correct model order. This result is in agreement with the findings of Yonekura et al. (2021), who show theoretically that BIC is strongly consistent for a general class of HMMs. The modified BIC (which we denote by BIC in the following) is defined as

$$\text{BIC} = \frac{1}{T} \{2 (L_T(\bar{\lambda}, \bar{\kappa}, \bar{K}; \mathbf{Y}) - L_T(\lambda, \kappa, K; \mathbf{Y})) + M \log(T)\} , \quad (6)$$

where $L_T(\lambda, \kappa, K; \mathbf{Y})$ denotes the estimated WCSS, $\mathbf{Y} \in \mathbb{R}^{T \times P}$ is the matrix of features, and λ , κ , and K are the values for the jump penalty, sparsity hyperparameter, and number of latent states, respectively. The terms $\bar{\lambda}$, $\bar{\kappa}$, and $\bar{K}$ are the SJM hyperparameters corresponding to the *saturated* model, i.e. the model obtained considering the complete set of features. While it is straightforward to set $\bar{\lambda} = 0$ and $\bar{\kappa} = \sqrt{P}$, it is not clear how to choose $\bar{K}$. Based on our experience, if the goal is to estimate a model with recurrent states, we suggest to not exceed $\bar{K} = 6$. Indeed, when $\bar{K}$ is too high, the modified BIC selects a large number of states, each one being visited only once. We define M in equation (6) by

$$M = K_0 |\mathcal{A}_0| + |\mathcal{A}_0| (K - K_0) + K_0 (|\mathcal{A}| - |\mathcal{A}_0|) + \sum_t \mathbb{I}_{s_t \neq s_{t-1}} , \quad (7)$$

where the three first terms come from a linear approximation of $K|\mathcal{A}|$ near the point $(K_0, |\mathcal{A}_0|)$. This expression penalizes for increasing values of K and $|\mathcal{A}|$, the number of latent states and active features. $|\mathcal{A}|$ indirectly depends on the hyperparameter κ , and it increases with increasing values of

κ . The last term in equation (7) counts the number of jumps across states and thereby depends indirectly on the jump penalty λ .

In practical applications, we suggest to select K_0 and $|\mathcal{A}_0|$ based on prior knowledge of the number of latent states and relevant features. In our empirical work, we set $|\mathcal{A}_0| = 150$ as we surmise that first and second order moments, correlations, volumes and momentum related features may be relevant, and these features are roughly 30 for each of the five cryptocurrencies.

The choice of K_0 is based primarily on qualitative properties. Selecting it too small will restrict the dynamics. Too large and the model does not generate persistent or even recurrent states. Rather, it segments the time series into separate blocks, thus negatively impacting model predictability. Koki et al. (2022) find that a non-homogeneous HMM with four states best describes BTC, ETH and XRP log-returns dynamics. In his comparative analysis, Bulla (2011) observes that considering an HMM with Student- t conditional distributions results into selecting a fewer number of regimes. In fact, compared to Gaussian conditional distributions, model estimation is less dependent on a few extreme observations that might cause the number of states to increase. Nystrup et al. (2020b) show that the JM is robust against distributional misspecification, similar to using an HMM with Student- t conditional distribution. Hence, we use $K_0 = 3$, reflecting our prior assumption of a modest number of states. In our setting, assuming an *a priori* number of states equal to 2, 3 or 4 does not significantly change the results.

3. ECONOMETRIC FEATURES

We construct a large set of features as potential candidates for explaining cryptocurrency returns. Many of these features have been proposed in the literature (see, Yae et al. (2022) for a survey), but some are new. We categorize them into three groups: financial market, sentiment, and crypto market-related features. Financial market features are based on information from the equity, fixed income, foreign exchange and commodity markets. Sentiment features proxy for investor attention toward the cryptocurrency markets. Crypto market-related features include prices and volumes of the cryptocurrencies as well as metrics related to the blockchain.

Some of our features require parameter choices before they are calculated, such as window length and the number of observations. This is addressed in

our empirical study by including multiple versions of the same variable computed for different parameters, from which the feature selection algorithm then can choose the most suitable appropriate features. Next, we describe the construction of the features in each group.

3.1. Financial Market Features. There is an ongoing debate whether cryptocurrencies constitute a separate asset class, with some market participants arguing that due fat-tails, high kurtosis and conditional volatility of their returns (Pele et al., 2021), they behave significantly differently than traditional assets. Nevertheless, it is natural to ask whether there exists some relationship between cryptocurrencies and traditional assets classes. Therefore, we construct a number of features based on time series from the equity, fixed income, foreign exchange and commodity markets, aimed at describing cross-asset dynamics between cryptocurrencies and traditional markets. In particular, we consider the following features: first differences of WTI oil prices; first differences of 10-year minus 3-months constant maturity Treasury yields (T10Y3M); log-returns of gold; log-returns of the S&P500; log-returns of NASDAQ; log-returns of EUR-USD; log-returns of JPY-USD; log-returns of CNY-USD; log-differences of VIX index; and *exponential moving averages* (EMAs) for log-differences of VIX index with half-lives $d = 1, 2, 7$, and 14 days. To proxy for possible comovements of cryptocurrencies relative to traditional asset classes (Selmi et al., 2018), we include *exponentially weighted linear and Gerber correlations*, denoted by ρ_d and g_d , (Gerber et al., 2022) of BTC log-returns with all other financial market features above with half-lives $d = 1, 2, 7$, and 14 days.

3.2. Sentiment Features. Cheah et al. (2020) suggest that investor sentiment such as public attention influence the crypto markets. Similarly, Aalborg et al. (2019) find that Google searches for BTC can predict BTC trading volume.

As proxies for public attention toward cryptocurrencies, we use the log-differences of the Google Trends indexes (GT) from the queries “bitcoin”, “ethereum”, “ripple”, “litecoin”, and “bitcoin cash”. We also add a second set of public attention-related features computed as exponentially weighted linear and Gerber correlations with half-lives $d = 1, 2, 7$, and 14 days of log-differences of Google Trend indexes and log-returns of each cryptocurrency.

3.3. Crypto Market-Related Features. This category cover features derived directly from cryptocurrency prices, trade volumes, and blockchain-related metrics. Log-differences of the total number of unique addresses with balance (AddWB) used on the blockchain aims at measuring the use value of a given coin. The number of addresses with balance is defined as the number of unique identifiers that serves as a virtual location where the coin can be sent. This metric differs from the total number of unique addresses in that it only counts wallets currently holding a particular coin, while the other one considers all addresses ever created. We use first differences of the total volume on chain (VOC), as a feature for the aggregate volume of transactions recorded on chain. Following Cong et al. (2021), we construct three value factors as the ratio between the total number of addresses and prices (AM), the number of addresses with balance and prices (UM), and the recorded volume on chain and prices (TM). We compute all the above mentioned features only for BTC, ETH, LTC and BCH due to data availability issues.

Hash rates refer to the amount of computing power used by the cryptocurrency network to process transactions and serves as a measure of the production cost of the mining process (Xiong et al., 2020). We include log-differences of hash rates (HR) for BTC and ETH.

To capture first and second moments of cryptocurrency returns, we include USD denominated daily log-returns, and EMAs of log-returns and volatilities with half-lives $d = 1, 2, 7$ and 14 days. To represent market activity, we use first differences of the logarithm of USD denominated trading volumes (V) and the corresponding EMAs with the same half-lives as above.

The results from several studies suggest that time series momentum is an important driver of crypto-returns (see, for example Liu et al. (2021), Liu et al. (2022), and Yae et al. (2022)). Therefore, we include several different momentum-based features in this study. Specifically, for each of the cryptocurrencies we consider the time series momentum signal (RF) of Moskowitz et al. (2012) which is based on time series regressions with a variable lag of l . In our empirical work we use $l = 1, 2, 7$ and 14 days. Moreover, taking inspiration from the technical analysis literature, we include the relative strength index (RSI) and the moving average converge-divergence minus signal (MACDS) indicator (Wilder, 1978; Appel, 2005). The MACDS and RSI are features that represent trend and reversal signals, respectively, and

are often used together to determine whether markets are in either a trending or range-bound condition. We apply the standard parameter choices when computing these features.

To proxy for illiquidity, we include the Amihud (2002) illiquidity measure (AMIHU), computed as the ratio of absolute daily log-returns and daily volumes for each coin.

Finally, we also construct exponentially weighted linear and Gerber correlations with half-lives $d = 1, 2, 7$ and 14 days of (a) BTC log-returns and log-returns of all other cryptocurrencies to obtain estimates of market betas, (b) log-returns and log-differences of trade volumes for each crypto, (c) BTC and ETH log-returns and log-differences of their corresponding hash rates, (d) BTC, ETH, LTC and BCH log-returns and their corresponding VOC and AddWB.

4. EMPIRICAL STUDY

4.1. Data. The study by Alexander et al. (2020) emphasize that for empirical analysis of cryptocurrency markets the choice of data sources is critical. In particular, they advise that researchers use trade data obtained from the crypto exchanges, rather than non-trade data from coin-ranking websites and other sources where data quality is significantly lower. Barucci et al. (2022) highlight that for intraday settings, cryptocurrencies quoted against BTC, ETH or Tether (USDT) are more liquidity and therefore tend to be more accurate than those quoted against the dollar. In fact, USDT facilitates trades in cryptocurrencies as fees are lower and no bank transfers are needed.

In the present work, we take the perspective of a USD denominated investor and therefore use USD prices and volumes.¹ Following Pennoni et al. (2021), we use price and trade volume data from the Crypto Asset Lab (CAL)², who collect data from crypto exchanges that satisfy their reliability and liquidity criteria. We are grateful to CAL for providing us

¹To ensure that the results of our study is not an artefact of possible differences in daily USD vs. USDT cryptocurrency quotations, we also perform our analysis with USDT denominated daily prices and volumes obtained from KuCoin. Due to data availability issues, the data for this comparative analysis covers a shorter time period, from March 7, 2019 through September 13, 2022. We observe no significant differences between the model estimated with USD or USDT denominated data. The results of this comparative analysis are available upon request.

²The Crypto Asset Lab is an independent lab established at the University of Milano-Bicocca; see <https://cryptoassetlab.diseade.unimib.it>

	Mean (%)	SD (%)
BTC	0.02	4.05
ETH	0.01	5.25
XRP	-0.09	6.03
LTC	-0.08	5.48
BCH	-0.18	6.24

TABLE 1. Daily unconditional means and standard deviations (SD) of the five cryptocurrency log-returns.

	BTC	ETH	XRP	LTC	BCH
BTC	1.00	-	-	-	-
ETH	0.84	1.00	-	-	-
XRP	0.65	0.70	1.00	-	-
LTC	0.81	0.84	0.71	1.00	-
BCH	0.78	0.81	0.68	0.83	1.00

TABLE 2. Unconditional correlation matrix of the five cryptocurrency log-returns.

with daily volume-weighted USD denominated prices and aggregate trade volumes, recorded at midnight UTC, from the Coinbase-pro, Poloniex, Bitstamp, Gemini, Bittrex, Kraken, and Bitflyer digital exchanges.

We obtain treasury constant maturity yields data from FRED³; gold and WTI oil prices, S&P500, NASDAQ and VIX levels, EUR-USD, JPY-USD and CNY-USD exchange rates from Bloomberg; hash rates, number of unique addresses with balance and volumes on chain from intotheblock.com; and online search trends for the five cryptocurrencies from Google Trends⁴.

Our dataset spans the time period from January 16, 2018 through September 13, 2022, with a total number of 1,702 daily observations. As traditional financial markets are only open during business days, their corresponding time series lack values during weekends. For convenience, we impute any missing data using the `mice` **R** package (Van Buuren et al., 2011). Table 1 and Table 2 provide the unconditional means, standard deviations and correlation matrix of the five cryptocurrency log-returns. From the different time series, we derive a total of 401 features, all of which are stationary (see Section 3 for a description of the construction of the features and Appendix A for a complete list).

³<https://fred.stlouisfed.org>

⁴<https://trends.google.com>

	State 1		State 2		State 3	
	Mean (%)	SD (%)	Mean (%)	SD (%)	Mean (%)	SD (%)
BTC	1.02	3.64	-0.26	3.15	-1.39	5.72
ETH	1.36	4.55	-0.29	4.23	-2.05	7.38
XRP	0.87	6.27	-0.26	4.80	-1.69	7.42
LTC	1.22	5.22	-0.47	4.17	-1.87	7.51
BCH	1.31	6.15	-0.46	4.59	-2.57	8.37

TABLE 3. State-conditional means and standard deviations (SD) of the five cryptocurrency log-returns obtained from the SJM model.

4.2. Results. We use the `Python` implementation of the SJM from Nystrup et al. (2021), available from their online supplementary material.⁵ The estimation of the SJM requires 3.1 seconds on a 16-core Intel i7-8750H with 16GB of RAM. For the remainder of this section, we discuss the results from the model.

Based on the BIC, we select the model with $K = 3$ states and with $\lambda = 5$, $\kappa = 4$. The daily state-conditional means and standard deviations of the returns are noticeably increasing from state 1 to state 3, as shown in Table 3. In Table 4, we observe that the correlations of the cryptocurrency log-returns increase from the first through the third state, where the correlations in the third state are remarkably high. This increase in correlations is consistent with the asymmetric correlation phenomena *within* equities and other asset classes, as well as *across* asset classes (see, for example Erb et al. (1994), De Bandt et al. (2000), and Cappiello et al. (2006)). Figure 2 depicts the decoded state sequence together with the cumulative log-returns of BTC, ETH, XRP, LTC and BCH. Throughout our sample of 1,702 observations, the model spends 38.9%, 41.8% and 19.3% of its time in each of the three different states, where each visit lasts for an average of 26.5, 17.8 and 16.4 days.

Together, the conditional statistics above suggest an interpretation of the states as distinct market regimes, where the first, second and third states represent a bull, neutral, and bear market regime, respectively. While the first regime (bull market) has positive average return and moderate volatility, the second regime (neutral market) is characterized by an average return slightly below zero under moderate volatility but with higher correlations

⁵<https://www.sciencedirect.com/science/article/pii/S0957417421009647#appSB>

State 1	BTC	ETH	XRP	LTC	BCH
BTC	1.00	-	-	-	-
ETH	0.72	1.00	-	-	-
XRP	0.50	0.57	1.00	-	-
LTC	0.70	0.73	0.58	1.00	-
BCH	0.63	0.67	0.54	0.73	1.00

State 2	BTC	ETH	XRP	LTC	BCH
BTC	1.00	-	-	-	-
ETH	0.84	1.000	-	-	-
XRP	0.62	0.66	1.00	-	-
LTC	0.83	0.86	0.67	1.00	-
BCH	0.81	0.81	0.65	0.84	1.00

State 3	BTC	ETH	XRP	LTC	BCH
BTC	1.00	-	-	-	-
ETH	0.92	1.00	-	-	-
XRP	0.81	0.87	1.00	-	-
LTC	0.89	0.92	0.89	1.00	-
BCH	0.89	0.90	0.86	0.91	1.00

TABLE 4. State-conditional correlations of the five cryptocurrency log-returns obtained from the SJM model.

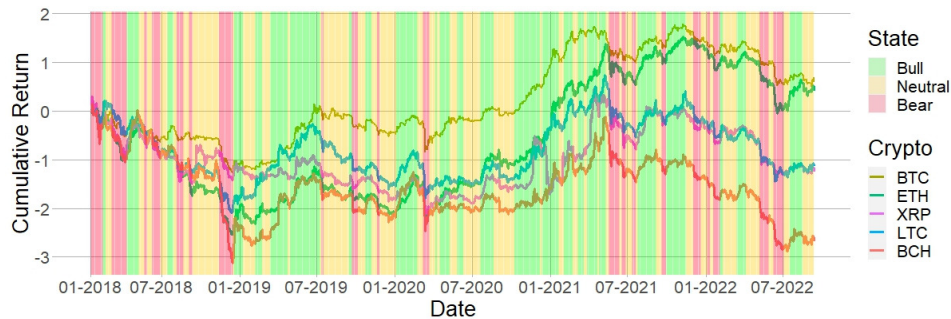


FIGURE 2. Cumulative log-returns of BTC, ETH, XRP, LTC, and BCH over the period January 2018 - September 2022, together with the state sequence obtained from the SJM in green (bull), yellow (neutral) and red (bear).

than the first regime. In contrast, the third regime (bear market) is associated with a significant average negative return and high volatility, approximately twice the magnitude of the volatilities observed in the two other regimes. In addition, cryptocurrency returns are highly correlated in the bear market regime, suggesting that there is little to no cross-sectional diversification during bad times.

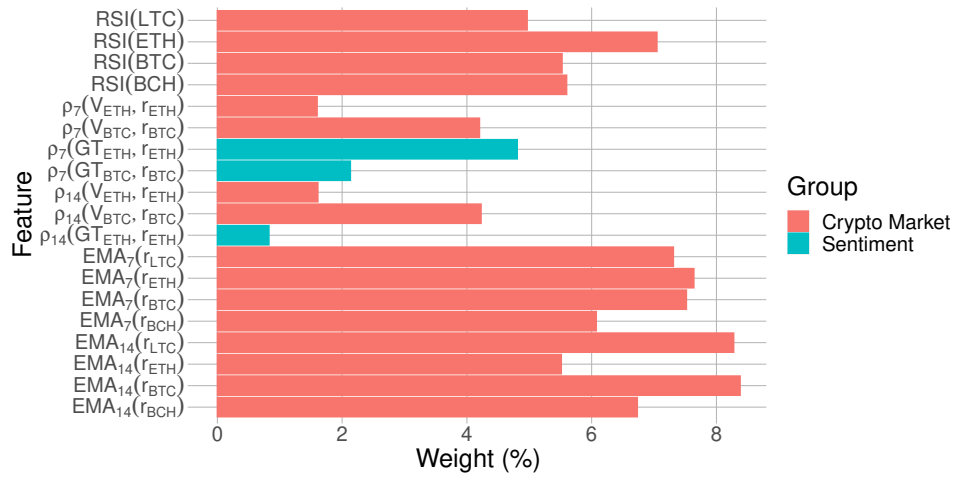


FIGURE 3. Estimated weights of the relevant features in the three-state model. The selected features are RSIs for BTC, ETH, LTC and BCH; 7- and 14-day exponentially weighted linear correlations of log-differences of volumes and log-returns for BTC and ETH; 7-day exponentially weighted linear correlations of log-differences of GT and log-returns for BTC and ETH; 14-day exponentially weighted linear correlations of log-differences of GT and log-returns for BTC; 7- and 14-day EMAs of log-returns of BTC, ETH, LTC and BCH. RSI, ρ_d and EMA_d denote the relative strength index, exponentially weighted linear correlation and exponential moving average with a half-life of d days, respectively.

Next, we turn to examining the features selected by the SJM. From the 401 features, the model selects 19 as being relevant (see, Figure 3 for a depiction of the feature weights of the relevant features). We observe that the 7- and 14-day EMAs of log-returns are the most relevant, with RSIs following thereafter. The state-conditional values of the relevant features are given in Table 5. These values are consistent with the bull (positive trend), neutral (range-bound) and bear (negative trend) regime interpretations above. The state-conditional 7- and 14-day EMAs for log-returns are similar in sign and magnitude to the state-conditional means and are consistent with the upward and downward momentum observed in the first and third regime.

The state-conditional RSI for BTC are 66.46, 45.03 and 32.58, consistent with the bull (positive trend), neutral (range-bound) and bear (negative trend) regime interpretations above. State-conditional RSIs for the other cryptocurrencies have similar magnitudes.

We observe that the 7-day exponentially weighted correlation of log-differences of Google Trend index and log-returns of BTC is relevant. The

Feature	Weight(%)	Bull	Neutral	Bear
RSI(LTC)	5.0	63.03	43.96	31.81
RSI(ETH)	7.0	66.68	45.03	31.61
RSI(BTC)	5.5	66.46	45.71	31.58
RSI(BCH)	5.6	62.55	43.55	28.60
$\rho_7(V_{ETH}, r_{ETH})$	1.6	0.21	-0.09	-0.32
$\rho_7(V_{BTC}, r_{BTC})$	4.2	0.24	-0.07	-0.36
$\rho_7(GT_{ETH}, r_{ETH})$	4.8	0.31	-0.08	-0.35
$\rho_7(GT_{BTC}, r_{BTC})$	2.1	0.26	-0.10	-0.38
$\rho_{14}(V_{ETH}, r_{ETH})$	1.6	0.22	-0.06	-0.29
$\rho_{14}(V_{BTC}, r_{BTC})$	4.2	0.18	-0.04	-0.29
$\rho_{14}(GT_{ETH}, r_{ETH})$	0.8	0.14	-0.08	-0.27
EMA ₇ (r_{ETH})	7.6	1.08%	-0.25%	1.50%
EMA ₇ (r_{LTC})	7.3	0.92%	-0.34%	-1.56%
EMA ₇ (r_{BTC})	7.5	0.80%	-0.15%	-1.18%
EMA ₇ (r_{BCH})	6.1	0.98%	-0.39%	-2.04%
EMA ₁₄ (r_{ETH})	5.5	0.78%	-0.17%	-1.00%
EMA ₁₄ (r_{LTC})	8.3	0.65%	-0.27%	-1.12%
EMA ₁₄ (r_{BTC})	8.4	0.60%	-0.10%	-0.86%
EMA ₁₄ (r_{BCH})	6.7	0.65%	-0.33%	-1.46%

TABLE 5. Estimated weights and state-conditional values of the selected features. RSI, ρ_d and EMA_d denote the relative strength index, exponentially weighted linear correlation and exponential moving average with a half-life of d days, respectively.

state-conditional values are equal to 0.26, -0.10 and -0.38 in the bull, neutral and bear market states, respectively. This suggests that public attention affects the evolution of crypto markets, especially during downward and upward market trends. The model selects the 7- and 14-day exponentially weighted correlation of log-differences of Google Trend index and log-returns of ETH, having state-conditional values similar to that of $\rho_7(GT_{BTC}, r_{BTC})$. Similarly, the 7- and 14-day exponentially weighted linear correlations of log-returns and log-differences of the BTC and ETH trade volumes are also selected. The state-conditional values in the bull, neutral and bear market states of the 7-day correlations for BTC are 0.24, -0.07 and -0.36 , respectively. The corresponding values for the 14-day correlations of BTC and for the 7- and 14-day correlations of ETH are similar in magnitude. Finally, we note the model does not select any features from the group of financial market features (cf. Section 3.1). In fact, the average state conditional correlations of BTC log-returns and each of the financial features are close to zero in all the three regimes.

4.3. Discussion. The SJM model distinguishes three distinct regimes driven by upward, downward and sideways trends, suggesting that time series momentum is a key driver of cryptocurrencies. Notably, the presence of time series momentum is well-established in traditional asset classes such as equity, currency, commodity, and fixed income markets (Moskowitz et al., 2012; Babu et al., 2020). More recently it has also been shown to be prevalent in the crypto markets (Cheah et al., 2020; Liu et al., 2021; Liu et al., 2022; Koki et al., 2022). Theories of sentiment in the behavioral finance literature suggests time series momentum may result from initial under-reaction followed by delayed over-reaction.⁶ Below we provide some support for this explanation and discuss how the interplay between institutional investors, who predominately act as market makers in these markets, and retail investors, who are holding positions longer, contribute to the trends.

While the interest of larger financial institutions in the crypto markets is growing, Karniol-Tambour et al. (2022) estimate that as of January 2022 only around 5% of Bitcoin is *held* by institutional investors. However, although institutions do not appear to have the largest share of holdings in the crypto markets, in the last few years they are generally believed to be the dominant players when it comes to trading volumes (e.g. market making). In 2021, institutions traded over one trillion dollars worth of cryptocurrencies on Coinbase, an increase from the 120 billion dollar trading volume the previous year, and more than twice the amount traded by retail investors (half a trillion dollars) (Vigna, 2022). As far as positive momentum, Auer et al. (2022) show that an increase in the price of Bitcoin causes a significant entry of new retail investors in the crypto markets who in turn drive up prices further, consistent with a positive feedback trading explanation (De Long et al., 1990). Using a dataset on client transactions and account balances of retail customers at a large German online bank, Hackethal et al. (2022) suggest that customers investing in cryptocurrencies and cryptocurrency structured retail products are likely to exhibit investing biases consistent with naive trend-chasing and overtrading behavior (Barber et al.,

⁶Under-reaction is the failure of markets to fully react to new information and can occur through several behavioral channels, including gradual dissemination of news (Hong et al., 1999), adherence to prior beliefs and cognitive biases such as anchoring (Barberis et al., 1998), or selling profitable assets too early while holding on to losing ones too long (Shefrin et al., 1985). Conversely, over-reaction is the tendency of markets reacting too strongly to new information and can result from excessive optimism and self-attribution biases (Daniel et al., 1998), herding behavior (Bikhchandani et al., 1992), positive feedback trading (De Long et al., 1990), or market sentiment (Baker et al., 2006).

2008). Additionally, Kogan et al. (2022) show that many retail investors actually end up following momentum strategies, whether they are aware of it or not, when investing in cryptocurrencies. The authors argue that this behavior is predominantly driven by retail investors holding on to their positions, even in periods of large price moves. In particular, they do not rebalance after prices increase or double up when prices decrease.

The features representing the interaction of cryptocurrency returns with public attention and trade volumes are also selected by our model. Their state-conditional values are positive in the bull regime, negative in the bear regime, and close to zero in the neutral regime, consistent with, for example, Bianchi et al. (2019) and Smales (2022).

Inspecting which groups of features are not selected by the SJM provides additional insight into the workings of crypto markets. Most importantly, features representing traditional asset classes are not helpful in explaining cryptocurrencies (see also Baur et al. (2018) and Bianchi (2020)) and neither is cryptocurrency return volatility-based features. That the latter are not selected is perhaps a bit surprising, especially as volatility-based features are some of the most important features for identifying regimes in the equity markets (Nystrup et al., 2020a; Nystrup et al., 2021). A possible reason for their non-inclusion is that their state-conditional values are about the same in the bull and neutral regimes, with each being about half of the corresponding values in the bear regime.

5. CONCLUSIONS

We employed the statistical sparse jump model to infer key features that drive the return dynamics of the largest cryptocurrencies. Our results suggest that a model with three states provides an intuitive interpretation of these markets corresponding to bull, neutral and bear market regimes. We found that first moments of returns (but not second moments), features representing trends and reversal signals drawn from the technical analysis literature, market activity and public attention have the strongest descriptive power. The features that we use for representing market activity and public attention are new and aid in explaining cryptocurrency returns in upward and downward market trends.

These findings have practical implications for trading and risk management in the crypto market. In particular, practitioners can use the identified features to distinguish upward and downward market trends, and detect when the market switches between different regimes.

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Conflict of interest. We have no conflicts of interest to disclose.

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APPENDIX A. FEATURE SET

Tag	Variable(s)	Transformation	Group
r_{BTC}	BTC log-ret	log-difference	Crypto Market-Related
r_{ETH}	ETH log-ret	log-difference	Crypto Market-Related
r_{XRP}	XRP log-ret	log-difference	Crypto Market-Related
r_{LTC}	LTC log-ret	log-difference	Crypto Market-Related
r_{BCH}	BCH log-ret	log-difference	Crypto Market-Related
V_{BTC}	BTC Volume	log-difference	Crypto Market-Related
V_{ETH}	ETH Volume	log difference	Crypto Market-Related
V_{XRP}	XRP Volume	log-difference	Crypto Market-Related
V_{LTC}	LTC Volume	log-difference	Crypto Market-Related
V_{BCH}	BCH Volume	log-difference	Crypto Market-Related
$EMA_1(r_{BTC})$	BTC log-ret	1-day EMA	Crypto Market-Related
$EMA_1(r_{ETH})$	ETH log-ret	1-day EMA	Crypto Market-Related
$EMA_1(r_{XRP})$	XRP log-ret	1-day EMA	Crypto Market-Related
$EMA_1(r_{LTC})$	LTC log-ret	1-day EMA	Crypto Market-Related
$EMA_1(r_{BCH})$	BCH log-ret	1-day EMA	Crypto Market-Related
$EMA_1(\sigma_{BTC})$	BTC log-ret	1-day EMA volatility	Crypto Market-Related
$EMA_1(\sigma_{ETH})$	ETH log-ret	1-day EMA volatility	Crypto Market-Related
$EMA_1(\sigma_{XRP})$	XRP log-ret	1-day EMA volatility	Crypto Market-Related
$EMA_1(\sigma_{LTC})$	LTC log-ret	1-day EMA volatility	Crypto Market-Related
$EMA_1(\sigma_{BCH})$	BCH log-ret	1-day EMA volatility	Crypto Market-Related
$EMA_2(r_{BTC})$	BTC log-ret	2-day EMA	Crypto Market-Related
$EMA_2(r_{ETH})$	ETH log-ret	2-day EMA	Crypto Market-Related
$EMA_2(r_{XRP})$	XRP log-ret	2-day EMA	Crypto Market-Related
$EMA_2(r_{LTC})$	LTC log-ret	2-day EMA	Crypto Market-Related
$EMA_2(r_{BCH})$	BCH log-ret	2-day EMA	Crypto Market-Related
$EMA_2(\sigma_{BTC})$	BTC log-ret	2-day EMA volatility	Crypto Market-Related
$EMA_2(\sigma_{ETH})$	ETH log-ret	2-day EMA volatility	Crypto Market-Related
$EMA_2(\sigma_{XRP})$	XRP log-ret	2-day EMA volatility	Crypto Market-Related
$EMA_2(\sigma_{LTC})$	LTC log-ret	2-day EMA volatility	Crypto Market-Related
$EMA_2(\sigma_{BCH})$	BCH log-ret	2-day EMA volatility	Crypto Market-Related
$EMA_7(r_{BTC})$	BTC log-ret	7-day EMA	Crypto Market-Related
$EMA_7(r_{ETH})$	ETH log-ret	7-day EMA	Crypto Market-Related
$EMA_7(r_{XRP})$	XRP log-ret	7-day EMA	Crypto Market-Related
$EMA_7(r_{LTC})$	LTC log-ret	7-day EMA	Crypto Market-Related
$EMA_7(r_{BCH})$	BCH log-ret	7-day EMA	Crypto Market-Related
$EMA_7(\sigma_{BTC})$	BTC log-ret	7-day EMA volatility	Crypto Market-Related
$EMA_7(\sigma_{ETH})$	ETH log-ret	7-day EMA volatility	Crypto Market-Related
$EMA_7(\sigma_{XRP})$	XRP log-ret	7-day EMA volatility	Crypto Market-Related
$EMA_7(\sigma_{LTC})$	LTC log-ret	7-day EMA volatility	Crypto Market-Related
$EMA_7(\sigma_{BCH})$	BCH log-ret	7-day EMA volatility	Crypto Market-Related
$EMA_{14}(r_{BTC})$	BTC log-ret	14-day EMA	Crypto Market-Related
$EMA_{14}(r_{ETH})$	ETH log-ret	14-day EMA	Crypto Market-Related
$EMA_{14}(r_{XRP})$	XRP log-ret	14-day EMA	Crypto Market-Related
$EMA_{14}(r_{LTC})$	LTC log-ret	14-day EMA	Crypto Market-Related
$EMA_{14}(r_{BCH})$	BCH log-ret	14-day EMA	Crypto Market-Related
$EMA_{14}(\sigma_{BTC})$	BTC log-ret	14-day EMA volatility	Crypto Market-Related
$EMA_{14}(\sigma_{ETH})$	ETH log-ret	14-day EMA volatility	Crypto Market-Related
$EMA_{14}(\sigma_{XRP})$	XRP log-ret	14-day EMA volatility	Crypto Market-Related
$EMA_{14}(\sigma_{LTC})$	LTC log-ret	14-day EMA volatility	Crypto Market-Related
$EMA_{14}(\sigma_{BCH})$	BCH log-ret	14-day EMA volatility	Crypto Market-Related
$EMA_1(V_{BTC})$	BTC Volume	1-day EMA Volume	Crypto Market-Related
$EMA_1(V_{ETH})$	ETH Volume	1-day EMA Volume	Crypto Market-Related
$EMA_1(V_{XRP})$	XRP Volume	1-day EMA Volume	Crypto Market-Related
$EMA_1(V_{LTC})$	LTC Volume	1-day EMA Volume	Crypto Market-Related
$EMA_1(V_{BCH})$	BCH Volume	1-day EMA Volume	Crypto Market-Related
$EMA_2(V_{BTC})$	BTC Volume	2-day EMA Volume	Crypto Market-Related
$EMA_2(V_{ETH})$	ETH Volume	2-day EMA Volume	Crypto Market-Related
$EMA_2(V_{XRP})$	XRP Volume	2-day EMA Volume	Crypto Market-Related
$EMA_2(V_{LTC})$	LTC Volume	2-day EMA Volume	Crypto Market-Related
$EMA_2(V_{BCH})$	BCH Volume	2-day EMA Volume	Crypto Market-Related
$EMA_7(V_{BTC})$	BTC Volume	7-day EMA Volume	Crypto Market-Related
$EMA_7(V_{ETH})$	ETH Volume	7-day EMA Volume	Crypto Market-Related
$EMA_7(V_{XRP})$	XRP Volume	7-day EMA Volume	Crypto Market-Related
$EMA_7(V_{LTC})$	LTC Volume	7-day EMA Volume	Crypto Market-Related
$EMA_7(V_{BCH})$	BCH Volume	7-day EMA Volume	Crypto Market-Related
$EMA_{14}(V_{BTC})$	BTC Volume	14-day EMA Volume	Crypto Market-Related
$EMA_{14}(V_{ETH})$	ETH Volume	14-day EMA Volume	Crypto Market-Related
$EMA_{14}(V_{XRP})$	XRP Volume	14-day EMA Volume	Crypto Market-Related
$EMA_{14}(V_{LTC})$	LTC Volume	14-day EMA Volume	Crypto Market-Related
$EMA_{14}(V_{BCH})$	BCH Volume	14-day EMA Volume	Crypto Market-Related
$\rho_1(V_{BTC}, r_{BTC})$	BTC log-ret BTC Volume	1-day EMA linear correlation	Crypto Market-Related
$\rho_1(V_{ETH}, r_{ETH})$	ETH log-ret ET Volume	1-day EMA linear correlation	Crypto Market-Related

Continued on next page

Tag	Variable(s)	Transformation	Group
$g7(r_{BCH}, r_{BTC})$	BTC log-ret BCH log-ret	7-day linear correlation	Crypto Market-Related
$g14(r_{ETH}, r_{BTC})$	BTC log-ret ETH log-ret	14-day linear correlation	Crypto Market-Related
$g14(r_{XRP}, r_{BTC})$	BTC log-ret XRP log-ret	14-day linear correlation	Crypto Market-Related
$g14(r_{LTC}, r_{BTC})$	BTC log-ret LTC log-ret	14-day linear correlation	Crypto Market-Related
$g14(r_{BCH}, r_{BTC})$	BTC log-ret BCH log-ret	14-day linear correlation	Crypto Market-Related
TM(BTC)	BTC on chain volume with balance BTC prices	first-difference of ratio	Crypto Market-Related
TM(ETH)	ETH on chain volume with balance ETH prices	first-difference of ratio	Crypto Market-Related
TM(LTC)	LTC on chain volume with balance LTC prices	first-difference of ratio	Crypto Market-Related
TM(BCH)	BCH number of total addresses with balance BCH prices	first-difference of ratio	Crypto Market-Related
AM(BTC)	BTC number of total addresses BTC prices	log-difference of ratio	Crypto Market-Related
AM(ETH)	ETH number of total addresses ETH prices	log-difference of ratio	Crypto Market-Related
AM(LTC)	LTC number of total addresses LTC prices	log-difference of ratio	Crypto Market-Related
AM(BCH)	BCH number of total addresses BCH prices	log-difference of ratio	Crypto Market-Related
UM(BTC)	BTC number of total addresses with balance BTC prices	log-difference of ratio	Crypto Market-Related
UM(ETH)	ETH number of total addresses with balance ETH prices	log-difference of ratio	Crypto Market-Related
UM(LTC)	LTC number of total addresses with balance LTC prices	log-difference of ratio	Crypto Market-Related
UM(BCH)	BCH number of total addresses with balance BCH prices	log-difference of ratio	Crypto Market-Related
AMIHUDBTC	BTC absolute log-ret BTC volume	ratio	Crypto Market-Related
AMIHUDETH	ETH absolute log-ret ETH volume	ratio	Crypto Market-Related
AMIHUDXRP	XRP absolute log-ret XRP volume	ratio	Crypto Market-Related
AMIHUDBTC	LTC absolute log-ret LTC volume	ratio	Crypto Market-Related
AMIHUDBCH	BCH absolute log-ret BCH volume	ratio	Crypto Market-Related
GTBTC	BTC Google Index	first-difference	Sentiment
GETH	ETH Google Index	first-difference	Sentiment
GTXRP	XRP Google Index	first-difference	Sentiment
GTBTC	LTC Google Index	first-difference	Sentiment
GTBCH	BCH Google Index	first-difference	Sentiment
$\rho_1(GTBTC, r_{BTC})$	BTC log-ret BTC Google	1-day EMA linear correlation	Sentiment
$\rho_1(GTETH, r_{ETH})$	ETH log-ret ETH Google	1-day EMA linear correlation	Sentiment
$\rho_1(GTXRP, r_{XRP})$	XRP log-ret XRP Google	1-day EMA linear correlation	Sentiment
$\rho_1(GTLTC, r_{LTC})$	LTC log-ret LTC Google	1-day EMA linear correlation	Sentiment
$\rho_1(GTBCH, r_{BCH})$	BCH log-ret BCH Google	1-day EMA linear correlation	Sentiment
$\rho_2(GTBTC, r_{BTC})$	BTC log-ret BTC Google	2-day EMA linear correlation	Sentiment
$\rho_2(GTETH, r_{ETH})$	ETH log-ret ETH Google	2-day EMA linear correlation	Sentiment
$\rho_2(GTXRP, r_{XRP})$	XRP log-ret XRP Google	2-day EMA linear correlation	Sentiment
$\rho_2(GTLTC, r_{LTC})$	LTC log-ret LTC Google	2-day EMA linear correlation	Sentiment
$\rho_2(GTBCH, r_{BCH})$	BCH log-ret BCH Google	2-day EMA linear correlation	Sentiment
$\rho_7(GTBTC, r_{BTC})$	BTC log-ret BTC Google	7-day EMA linear correlation	Sentiment
$\rho_7(GTETH, r_{ETH})$	ETH log-ret ETH Google	7-day EMA linear correlation	Sentiment
$\rho_7(GTXRP, r_{XRP})$	XRP log-ret XRP Google	7-day EMA linear correlation	Sentiment
$\rho_7(GTLTC, r_{LTC})$	LTC log-ret LTC Google	7-day EMA linear correlation	Sentiment
$\rho_7(GTBCH, r_{BCH})$	BCH log-ret BCH Google	7-day EMA linear correlation	Sentiment
$\rho_{14}(GTBTC, r_{BTC})$	BTC log-ret BTC Google	14-day EMA linear correlation	Sentiment
$\rho_{14}(GTETH, r_{ETH})$	ETH log-ret ETH Google	14-day EMA linear correlation	Sentiment
$\rho_{14}(GTXRP, r_{XRP})$	XRP log-ret XRP Google	14-day EMA linear correlation	Sentiment
$\rho_{14}(GTLTC, r_{LTC})$	LTC log-ret LTC Google	14-day EMA linear correlation	Sentiment
$\rho_{14}(GTBCH, r_{BCH})$	BCH log-ret BCH Google	14-day EMA linear correlation	Sentiment
$g_1(GTBTC, r_{BTC})$	BTC log-ret BTC Google	1-day EMA Gerber correlation	Sentiment
$g_1(GTETH, r_{ETH})$	ETH log-ret ETH Google	1-day EMA Gerber correlation	Sentiment
$g_1(GTXRP, r_{XRP})$	XRP log-ret XRP Google	1-day EMA Gerber correlation	Sentiment
$g_1(GTLTC, r_{LTC})$	LTC log-ret LTC Google	1-day EMA Gerber correlation	Sentiment
$g_1(GTBCH, r_{BCH})$	BCH log-ret BCH Google	1-day EMA Gerber correlation	Sentiment
$g_2(GTBTC, r_{BTC})$	BTC log-ret BTC Google	2-day EMA Gerber correlation	Sentiment
$g_2(GTETH, r_{ETH})$	ETH log-ret ETH Google	2-day EMA Gerber correlation	Sentiment
$g_2(GTXRP, r_{XRP})$	XRP log-ret XRP Google	2-day EMA Gerber correlation	Sentiment
$g_2(GTLTC, r_{LTC})$	LTC log-ret LTC Google	2-day EMA Gerber correlation	Sentiment
$g_2(GTBCH, r_{BCH})$	BCH log-ret BCH Google	2-day EMA Gerber correlation	Sentiment
$g_7(GTBTC, r_{BTC})$	BTC log-ret BTC Google	7-day EMA Gerber correlation	Sentiment
$g_7(GTETH, r_{ETH})$	ETH log-ret ETH Google	7-day EMA Gerber correlation	Sentiment
$g_7(GTXRP, r_{XRP})$	XRP log-ret XRP Google	7-day EMA Gerber correlation	Sentiment
$g_7(GTLTC, r_{LTC})$	LTC log-ret LTC Google	7-day EMA Gerber correlation	Sentiment

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Tag	Variable(s)	Transformation	Group
$g_7(GT_{BCH}, r_{BCH})$	BCH log-ret BCH Google	7-day EMA Gerber correlation	Sentiment
$g_{14}(GT_{BTC}, r_{BTC})$	BTC log-ret BTC Google	14-day EMA Gerber correlation	Sentiment
$g_{14}(GT_{ETH}, r_{ETH})$	ETH log-ret ET Google	14-day EMA Gerber correlation	Sentiment
$g_{14}(GT_{XRP}, r_{XRP})$	XRP log-ret XRP Google	14-day EMA Gerber correlation	Sentiment
$g_{14}(GT_{LTC}, r_{LTC})$	LTC log-ret LTC Google	14-day EMA Gerber correlation	Sentiment
$g_{14}(GT_{BCH}, r_{BCH})$	BCH log-ret BCH Google	14-day EMA Gerber correlation	Sentiment
GOLD	Gold log-ret	log-difference	Financial Market
EURUSD	EURUSD log-ret	log-difference	Financial Market
JPYUSD	JPYUSD log-ret	log-difference	Financial Market
CNYUSD	CNYUSD log-ret	log-difference	Financial Market
VIX	VIX log-ret	log-difference	Financial Market
SP500	SP500 log-ret	log-difference	Financial Market
NASDAQ	NASDAQ log-ret	log-difference	Financial Market
T10Y3M	10 years minus 3 months US treasury yields	first difference	Financial Market
WTI	WTI	first-difference	Financial Market
$\rho_1(GOLD, r_{BTC})$	BTC log-ret gold log-ret	1-day EMA Gerber correlation	Financial Market
$\rho_2(GOLD, r_{BTC})$	BTC log-ret gold log-ret	2-day EMA Gerber correlation	Financial Market
$\rho_7(GOLD, r_{BTC})$	BTC log-ret gold log-ret	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(GOLD, r_{BTC})$	BTC log-ret gold log-ret	14-day EMA Gerber correlation	Financial Market
$g_1(GOLD, r_{BTC})$	BTC log-ret gold log-ret	1-day EMA Gerber correlation	Financial Market
$g_2(GOLD, r_{BTC})$	BTC log-ret gold log-ret	2-day EMA Gerber correlation	Financial Market
$g_7(GOLD, r_{BTC})$	BTC log-ret gold log-ret	7-day EMA Gerber correlation	Financial Market
$g_{14}(GOLD, r_{BTC})$	BTC log-ret gold log-ret	14-day EMA Gerber correlation	Financial Market
$\rho_1(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	1-day EMA Gerber correlation	Financial Market
$\rho_2(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	2-day EMA Gerber correlation	Financial Market
$\rho_7(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	14-day EMA Gerber correlation	Financial Market
$g_1(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	1-day EMA Gerber correlation	Financial Market
$g_2(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	2-day EMA Gerber correlation	Financial Market
$g_7(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	7-day EMA Gerber correlation	Financial Market
$g_{14}(EURUSD, r_{BTC})$	BTC log-ret EURUSD log-ret	14-day EMA Gerber correlation	Financial Market
$\rho_1(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	1-day EMA Gerber correlation	Financial Market
$\rho_2(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	2-day EMA Gerber correlation	Financial Market
$\rho_7(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	14-day EMA Gerber correlation	Financial Market
$g_1(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	1-day EMA Gerber correlation	Financial Market
$g_2(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	2-day EMA Gerber correlation	Financial Market
$g_7(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	7-day EMA Gerber correlation	Financial Market
$g_{14}(JPYUSD, r_{BTC})$	BTC log-ret JPYUSD log-ret	14-day EMA Gerber correlation	Financial Market
$\rho_1(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	1-day EMA Gerber correlation	Financial Market
$\rho_2(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	2-day EMA Gerber correlation	Financial Market
$\rho_7(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	14-day EMA Gerber correlation	Financial Market
$g_1(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	1-day EMA Gerber correlation	Financial Market
$g_2(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	2-day EMA Gerber correlation	Financial Market
$g_7(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	7-day EMA Gerber correlation	Financial Market
$g_{14}(CNYUSD, r_{BTC})$	BTC log-ret CNYUSD log-ret	14-day EMA Gerber correlation	Financial Market
$\rho_1(VIX, r_{BTC})$	BTC log-ret VIX log-ret	1-day EMA Gerber correlation	Financial Market
$\rho_2(VIX, r_{BTC})$	BTC log-ret VIX log-ret	2-day EMA Gerber correlation	Financial Market
$\rho_7(VIX, r_{BTC})$	BTC log-ret VIX log-ret	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(VIX, r_{BTC})$	BTC log-ret VIX log-ret	14-day EMA Gerber correlation	Financial Market
$g_1(VIX, r_{BTC})$	BTC log-ret VIX log-ret	1-day EMA Gerber correlation	Financial Market
$g_2(VIX, r_{BTC})$	BTC log-ret VIX log-ret	2-day EMA Gerber correlation	Financial Market
$g_7(VIX, r_{BTC})$	BTC log-ret VIX log-ret	7-day EMA Gerber correlation	Financial Market
$g_{14}(VIX, r_{BTC})$	BTC log-ret VIX log-ret	14-day EMA Gerber correlation	Financial Market
$\rho_1(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	1-day EMA Gerber correlation	Financial Market
$\rho_2(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	2-day EMA Gerber correlation	Financial Market
$\rho_7(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	14-day EMA Gerber correlation	Financial Market
$g_1(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	1-day EMA Gerber correlation	Financial Market
$g_2(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	2-day EMA Gerber correlation	Financial Market
$g_7(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	7-day EMA Gerber correlation	Financial Market
$g_{14}(SP500, r_{BTC})$	BTC log-ret SP500 log-ret	14-day EMA Gerber correlation	Financial Market
$\rho_1(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	1-day EMA Gerber correlation	Financial Market
$\rho_2(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	2-day EMA Gerber correlation	Financial Market
$\rho_7(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	14-day EMA Gerber correlation	Financial Market
$g_1(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	1-day EMA Gerber correlation	Financial Market
$g_2(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	2-day EMA Gerber correlation	Financial Market
$g_7(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	7-day EMA Gerber correlation	Financial Market
$g_{14}(NASDAQ, r_{BTC})$	BTC log-ret NASDAQ log-ret	14-day EMA Gerber correlation	Financial Market
$\rho_1(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	1-day EMA Gerber correlation	Financial Market

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Tag	Variable(s)	Transformation	Group
$\rho_2(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	2-day EMA Gerber correlation	Financial Market
$\rho_7(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	14-day EMA Gerber correlation	Financial Market
$g_1(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	1-day EMA Gerber correlation	Financial Market
$g_2(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	2-day EMA Gerber correlation	Financial Market
$g_7(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	7-day EMA Gerber correlation	Financial Market
$g_{14}(T10Y3M, r_{BTC})$	BTC log-ret T10Y3M	14-day EMA Gerber correlation	Financial Market
$\rho_1(WTI, r_{BTC})$	BTC log-ret WTI	1-day EMA Gerber correlation	Financial Market
$\rho_2(WTI, r_{BTC})$	BTC log-ret WTI	2-day EMA Gerber correlation	Financial Market
$\rho_7(WTI, r_{BTC})$	BTC log-ret WTI	7-day EMA Gerber correlation	Financial Market
$\rho_{14}(WTI, r_{BTC})$	BTC log-ret WTI	14-day EMA Gerber correlation	Financial Market
$g_1(WTI, r_{BTC})$	BTC log-ret WTI	1-day EMA Gerber correlation	Financial Market
$g_2(WTI, r_{BTC})$	BTC log-ret WTI	2-day EMA Gerber correlation	Financial Market
$g_7(WTI, r_{BTC})$	BTC log-ret WTI	7-day EMA Gerber correlation	Financial Market
$g_{14}(WTI, r_{BTC})$	BTC log-ret WTI	14-day EMA Gerber correlation	Financial Market
$EMA_1(VIX)$	VIX log-ret	1-day EMA	Financial Market
$EMA_2(VIX)$	VIX log-ret	2-day EMA	Financial Market
$EMA_7(VIX)$	VIX log-ret	7-day EMA	Financial Market
$EMA_{14}(VIX)$	VIX log-ret	14-day EMA	Financial Market